

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – CODING GOLF COMPETITION

Course Code:	CSA0305	Course Title:	Data Structures
CO Assessed:	CO4 – Articulation (BL3)	Assessment:	Assessment Tool 1 – Coding Golf Competition
Weightage:	35% of CIA	Total Marks:	100
CO4 Statement:	Apply hashing strategies for efficient data storage and sorting algorithms for arrangement of data.		
Student Name:	M.Deepika	Reg. No.:	192372296
Section / Batch:	2023	Date:	17/08/2024

Code of Conduct

I ____M.Deepika____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the Student: ____deepika.m____

Instructions

- Log in to the assigned coding platform - Code chef (<https://www.codechef.com/skill-test/c-languagesa>). Total: 100 Marks.
- You can only attempt this assessment once. Also, you will not be able to pause the assessment after starting.
- You are not allowed to switch tabs during the assessment, otherwise you will be blocked from the assessment.
- You will get a detailed report on your performance at the end of the assessment.

Section / Topic	Focus	Q Nos	Marks
-----------------	-------	-------	-------

Data structures	Code Chef	28	100
GRAND TOTAL:		100 Marks	

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20	Accurately interprets all problem requirements and constraints before coding.	Understands most requirements with minor omissions.	Partially understands the problem; misses some constraints.	Misinterprets the problem or ignores key requirements.
Correctness of Solution	30	Program compiles successfully and passes all hidden and visible test cases.	Program passes most test cases with minor errors.	Program passes some test cases but has logical issues.	Program fails to compile or produces incorrect results.
Data Structure & Algorithm Selection	20	Selects the most appropriate data structure and algorithm with an optimized solution.	Chooses an appropriate solution with minor inefficiencies.	Uses a workable but less efficient approach.	Uses an inappropriate data structure or algorithm.
Code Quality & Documentation	20	Well-structured, modular, readable, and properly indented code with meaningful identifiers.	Readable code with minor formatting issues.	Basic coding style with inconsistent formatting.	Poorly organized code that is difficult to understand.
Time & Space Optimization	10	Demonstrates excellent optimization and correctly analyses time and space complexity.	Good optimization with minor inefficiencies.	Limited optimization and incomplete complexity analysis.	No optimization and incorrect or missing complexity analysis.
Faculty In-charge		Course Coordinator		Program Director	
Signature: _____		Signature: _____		Signature: _____	
Date: _____		Date: _____		Date: _____	

codechef.com/practice/course/cpp/LPCPAS01/problems/LCPPAS01

Prev module 1 out of 10 Next

Statement AI Help

Code Output - MCQ

What does the following code snippet do?

```
#include <bits/stdc++.h>
using namespace std;

int main() {
    cout << 21 + 40;
    return 0;
}
```

Did you like the problem?
972 users found this helpful

Select one of the following options:

- ☐ Reads two numbers from the console
- ☒ Adds two numbers and prints the result
- ☐ Multiplies two numbers and prints the result
- ☐ None of the above

✓ Perfect answer!

See Answer Hide Solution

Submit Next

codechef.com/practice/course/cpp/LPCPAS01/problems/LCPPAS02

Prev 2 out of 10 Next

Statement AI Help

Print Coding, Chef!

How would you print the below text?

```
Coding, Chef!
```

Did you like the problem?
237 users found this helpful

Select one of the following options:

- ☒ cout << "Coding, Chef!";
- ☐ cout << Coding, Chef!;
- ☐ cout << 'Coding, Chef!';
- ☐ cout << Coding << Chef;

✓ Excellent work!

See Answer Hide Solution

Submit Next

codechef.com/practice/course/cpp/LPCPAS01/problems/LCPPAS03

Statement AI Help 3 out of 10

Identify Correct Syntax

What is the correct syntax for printing the number 5 in C++?

Did you like the problem?
167 users found this helpful

Select one of the following options:

- ☐ print(5);
- ☐ display(5);
- ☐ console.log(5);
- ☒ cout << 5;

✓ Awesome, you nailed it!

See Answer Hide Solution

Submit Next

30°C Mostly sunny

codechef.com/practice/course/cpp/LPCPAS01/problems/LCPPAS04

Statement AI Help 4 out of 10

Print difference of 10 and 3

How do you print the result of difference between 10 and 3?

Did you like the problem?
113 users found this helpful

Select one of the following options:

- ☒ cout << 10 - 3;
- ☐ cout << diff(10 - 3);
- ☐ diff(10 - 3)
- ☐ cout << 10 - 3

✓ Excellent work!

See Answer Hide Solution

Submit Next

30°C Mostly sunny

codechef.com/practice/course/cpp/LCPPAS01/problems/LCPPAS05

Statement AI Help

5 out of 10

Correct Syntax for Header
Identify the correct syntax from the options below.

Did you like the problem?
149 users found this helpful

Select one of the following options:

- ☐ #include <iostream>;
- ☐ ?include <iostream>;
- ☒ #include <iostream>
- ☐ #include <iostream>;

Great job, keep it up!

See Answer Hide Solution ^

Submit Next

codechef.com/practice/course/cpp/LCPPAS01/problems/LCPPAS06

Statement AI Help

6 out of 10

Print String num
How would you print the string stored in variable named `num`?

Did you like the problem?
172 users found this helpful

Select one of the following options:

- ☒ `cout << num;`
- ☐ `cout << "num";`
- ☐ `cout << value(num);`
- ☐ `cout << get(num);`

Excellent work!

See Answer Hide Solution ^

Submit Next

codechef.com/practice/course/cpp/LPCPAS01/problems/LCPPAS07

7 out of 10

Statement AI Help

Print 6 divided by 2
How can you print the result of 6 divided by 2?

Did you like the problem?
217 users found this helpful

Select one of the following options:

- ☒ `cout << 6 / 2;`
- ☐ `div(6, 2);`
- ☐ `cout << div(6, 2) ;`
- ☐ None of the above

✓ Awesome, you nailed it!

See Answer Hide Solution ^

Submit Next

codechef.com/practice/course/cpp/LPCPAS01/problems/LCPPAS08

8 out of 10

Statement AI Help

Identify Incorrect Syntax
Identify the incorrect syntax from the given options.

Did you like the problem?
297 users found this helpful

Select one of the following options:

- ☐ `cout << "Hello, World";`
- ☐ `cout << 5 + 3 / 2;`
- ☒ `cout << "Hello, World";`
- ☐ `cout << "5 + 3";`

✓ Hooray, you did it!

See Answer Hide Solution ^

Submit Next

codechef.com/practice/course/cpp/LPCPAS01/problems/LCPPAS09

Statement Submissions Solution AI Help C++ 9 out of 10

Switch to AI Tutor Mode

Print 108 using 9 and 12

Print the number 108 using arithmetic operations on the numbers 9 and 12.

Sample 1:

Input	Output
	108

Did you like the problem?
338 users found this helpful

Sample Input

Your Output

108

✓ You got it right! [AI Review?](#)

Sub-Task	Task #	Result (time)
1	0	Correct (0.00)

Subtask Score: 100% Result - Correct

Total Score = 100%

[Visualize Code](#) [Submit](#) [Next](#)

30°C Mostly sunny 9:15 AM 26-08-2024

codechef.com/practice/course/cpp/LPCPAS01/problems/LCPPAS10

Statement Submissions Solution AI Help C++ 10 out of 10

Switch to AI Tutor Mode

Print Learn Coding on CodeChef

Print "Learn Coding on CodeChef" to the console.

Sample 1:

Input	Output
	Learn Coding on CodeChef

Did you like the problem?
207 users found this helpful

Sample Input

Your Output

Learn Coding on CodeChef

✓ You got it right! [AI Review?](#)

Sub-Task	Task #	Result (time)
1	0	Correct (0.00)

Subtask Score: 100% Result - Correct

Total Score = 100%

[Visualize Code](#) [Submit](#) [Next](#)

30°C Mostly sunny 9:16 AM 26-08-2024